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Faculty of Informatics

Bioinformatics

Homework #2

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1 BLAST exercise

Given the two sequence:

C C Y E K R R K H Y C Q H C N Q W W V E W

and

C C L V I L P G H C D D I E W

and the following parameters:

- w=3
- acceptance = 25
- extension = 20
- matrix = BLOSUM50
- gap open = -5
- gap extend = -1

we need to provide a possible alignment following the BLAST procedure.

1.1 Build w-mer matrix

The first step to align following the BLAST procedure need to create the w-mer matrix, that is the matrix of the consecutive triplets (since we have w=3) of the two sequence give in the exercise. To fill the matrix, we must use the value inside the Blosum50 matrix, for example, the first triplet for the first sequence is:

C C Y

while the first triplet for the second sequence is

C C L

since in the Blosum50 matrix the a match of the Cysteine with the Cysteine have value 13 and the mismatch of Tyrosine and Leucine have a value of -1, the sum of $13 + 13 - 1 = 25$ and so we put this value in the first cell. We continue completing the matrix selecting the second triplet of the first sequence (C Y E) and the first triplet of the second sequence (again C C L) and evaluate the sum of the match/mismatch using the Blosum50 matrix (

$$13 - 3 - 3 = 7$$

) and we put this value in the second cell. And so on...
We show below the matrix completed

	CCY	CYE	YEK	EKR	KRR	RRK	RKH	KHY	HYC	YCQ	CQH	QHC	HCN	CNQ	NQW	QWW	WWV	WVE	VEW
CCL	25	7	-9	-9	-10	-11	-10	-7	-8	8	7	-8	6	9	-7	-10	-9	-9	-6
CLV	10	9	-9	-9	-9	-10	-11	-7	-5	-8	7	-7	-8	6	-7	-8	-2	-7	-7
LVI	-4	-7	-7	-10	-10	-9	-10	-8	-6	-5	-9	-8	-7	-8	-10	-8	-1	-1	-5
VIL	-4	-5	-8	-9	-10	-10	-9	-8	-7	-5	-7	-9	-10	-6	-8	-8	-5	-2	-1
ILP	-7	-4	-5	-10	-9	-8	-9	-9	-9	-4	-6	-10	-8	-7	-9	-9	-8	-3	-3
LPG	-9	-8	-4	-7	-9	-8	-6	-8	-9	-7	-5	-7	-7	-6	-8	-9	-10	-8	-3
PGH	-5	-7	-6	-3	-4	-6	5	-1	-8	-5	4	-6	-4	-3	-7	-7	-11	-8	-9
GHC	-9	-4	-6	-7	-6	-6	-6	5	13	-9	-5	21	-7	-5	-4	-10	-7	-10	-9
HCD	7	-4	-2	-5	-6	-5	-4	-6	3	15	-7	-6	25	-5	-7	-9	-12	-2	-12
CDD	6	12	-2	-6	-7	-7	-6	-7	-10	-7	12	-8	-5	15	-7	-13	-14	-7	-4
DDI	-9	-11	-4	-3	-7	-7	-7	-3	-6	-10	-8	-3	-8	-5	-1	-8	-6	-13	-5
DIE	-8	1	-6	-1	-5	-5	-5	-7	-5	-3	-7	-7	-3	-5	-4	-6	-11	5	-11
IEW	-3	-7	2	-6	-6	-7	-6	-1	-11	-5	-3	-8	-11	-3	14	9	-9	-9	25

Figure 1

We then use the parameter of the acceptance to select only the cells that have a value $\geq$ than that (in this case 25) to find the HSP's.

	CCY	CYE	YEK	EKR	KRR	RRK	RKH	KHY	HYC	YCQ	CQH	QHC	HCN	CNQ	NQW	QWW	WWV	WVE	VEW
CCL	25	7	-9	-9	-10	-11	-10	-7	-8	8	7	-8	6	9	-7	-10	-9	-9	-6
CLV	10	9	-9	-9	-9	-10	-11	-7	-5	-8	7	-7	-8	6	-7	-8	-2	-7	-7
LVI	-4	-7	-7	-10	-10	-9	-10	-8	-6	-5	-9	-8	-7	-8	-10	-8	-1	-1	-5
VIL	-4	-5	-8	-9	-10	-10	-9	-8	-7	-5	-7	-9	-10	-6	-8	-8	-5	-2	-1
ILP	-7	-4	-5	-10	-9	-8	-9	-9	-9	-4	-6	-10	-8	-7	-9	-9	-8	-3	-3
LPG	-9	-8	-4	-7	-9	-8	-6	-8	-9	-7	-5	-7	-7	-6	-8	-9	-10	-8	-3
PGH	-5	-7	-6	-3	-4	-6	5	-1	-8	-5	4	-6	-4	-3	-7	-7	-11	-8	-9
GHC	-9	-4	-6	-7	-6	-6	-6	5	13	-9	-5	21	-7	-5	-4	-10	-7	-10	-9
HCD	7	-4	-2	-5	-6	-5	-4	-6	3	15	-7	-6	25	-5	-7	-9	-12	-2	-12
CDD	6	12	-2	-6	-7	-7	-6	-7	-10	-7	12	-8	-5	15	-7	-13	-14	-7	-4
DDI	-9	-11	-4	-3	-7	-7	-7	-3	-6	-10	-8	-3	-8	-5	-1	-8	-6	-13	-5
DIE	-8	1	-6	-1	-5	-5	-5	-7	-5	-3	-7	-7	-3	-5	-4	-6	-11	5	-11
IEW	-3	-7	2	-6	-6	-7	-6	-1	-11	-5	-3	-8	-11	-3	14	9	-9	-9	25

Figure 2

1.2 Extend the best HSP's

The next step is to extend the HSP's usin the parameter of the extension, so that we pick the triplets selected and we extend the sequences to both sides (or just one side if are the first ones or the last ones) stopping the extension when the score (again given by the Blosum50 matrix) is below the extension threshold (in this case 20).
For the first HSP we have

C	C	Y	E	K
C	C	L	V	I
	25		22	19

Were since we have -3 as value for the mismatch of Glutamate (E) and Valine (V) the score after the first extension is 22, which is higher than 20 so we continue with the second extension, and then since also the mismatch of Lysine (K) and Isoleucine (I) have value -3, the score become 19, which is lower tha 20, so we stop here and we only keep:

C	C	Y	E
C	C	L	V

For the second HSP we have:

C	Q	H	C	N	Q	W	W
P	G	H	C	D	D	I	E
19	23		25		25	22	19

In this case since is at the center of the sequence we must extend for both sides and we stop when we reach a value lower than 20. The final result is:

Q	H	C	N	Q	W
G	H	C	D	D	I

For the last HSP we have

W	W	V	E	W
D	D	I	E	W
15	20		25	

Were since are the last triplets we extend on the left and we again we stop when the score is lower than 20. So we have

W	V	E	W
D	I	E	W

1.3 Connect the final HSP and traceback

After extended the HSP, we create the DP matrix, and we start filling it by the extended HSP, putting in each cell of match/mismatch the value of the Blosum50 matrix, so that we have:

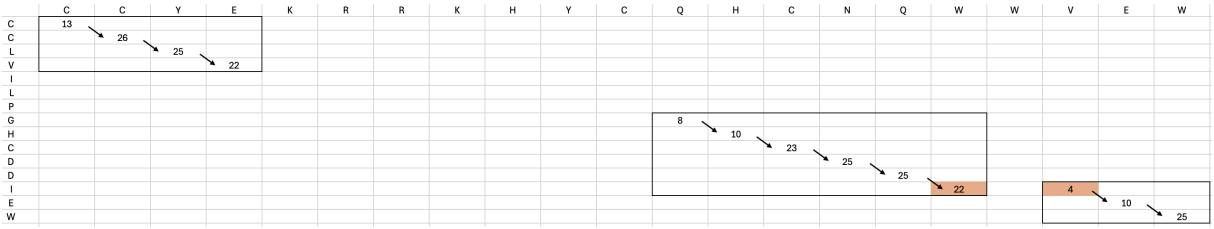

Figure 3

end the three squares represent the extended HSP. Now, what we should do is connect the three squares in the matrix using the Needleman Wunsch algorithm, but as we can see there are two overlapping match/mismatch (the two orange cells). In this case we can both select during the traceback the mismatch of Tryptophan (W) and Isoleucine (I) or the mismatch of Valine (V) and Isoleucine (I). To avoid this problem, we put a square that contain both the previous and the next amino acid to the ones that coincide, and perform the Needleman Wunsch algorithm in that point to find the correct traceback. We ends up with:

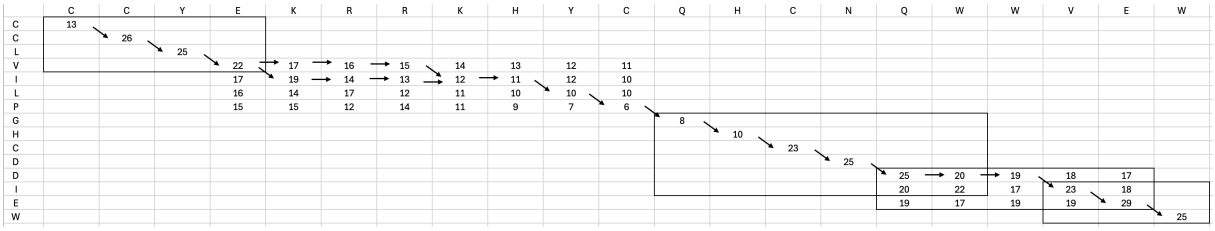

Figure 4

As we can see there are two possible trace back. The first is:

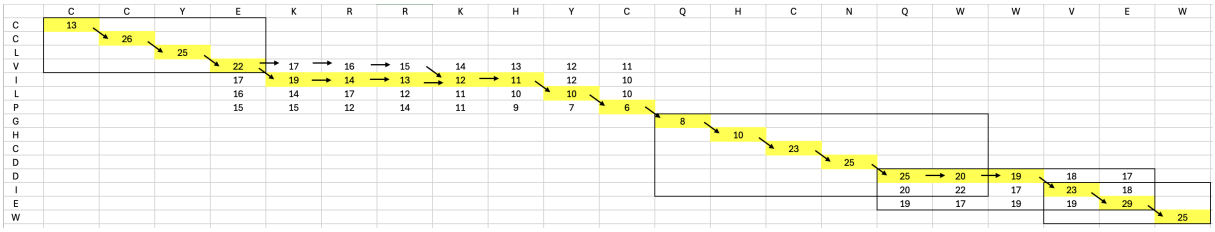

Figure 5

while the second one is:

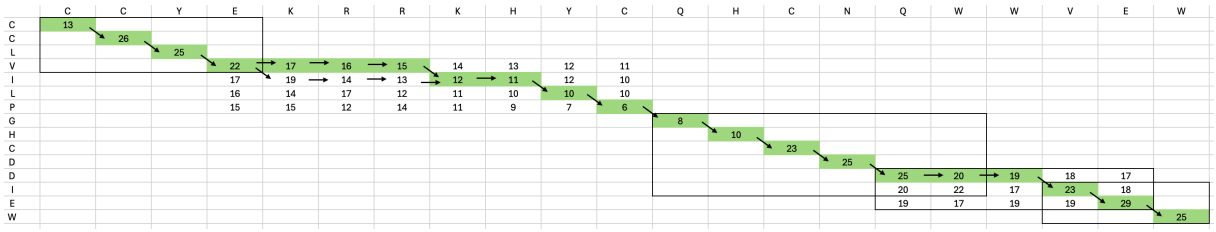

Figure 6

1.4 Alignment and global score

Finally, we take our traceback(s) and we perform the alignment and we evaluate the global score:

Alignment 1:

C	C	Y	E	K	R	R	K	H	Y	C	Q	H	C	N	Q	W	W	V	E	W
C	C	L	V	I	–	–	–	–	L	P	G	H	C	D	D	–	–	I	E	W
+13	+13	-1	-3	-3	-5	-1	-1	-1	-1	-4	-2	+10	+13	+2	+0	-5	-1	+4	+6	+15

Total score: +48

Alignment 2:

C	C	Y	E	K	R	R	K	H	Y	C	Q	H	C	N	Q	W	W	V	E	W
C	C	L	V	–	–	–	I	–	L	P	G	H	C	D	D	–	–	I	E	W
+13	+13	-1	-3	-5	-1	-1	-3	-5	-1	-4	-2	+10	+13	+2	+0	-5	-1	+4	+6	+15

Total score: +44

Since the first alignment is the one with the highest total score, we keep this as final alignment.

2 PSSM exercise

2.1 Point A

We need to arrive to the PSSM matrix for these sequences (aligned):

- RAT$\longrightarrow$ AGCAGAGAGTCAGTGAATACAGTGG
- CAT$\longrightarrow$ ATCTCCAGCCCCCAGGGGCCGGCGG
- DOG$\longrightarrow$ CACATTTCCTGAGGTGGGTCCTGTG
- DEER$\longrightarrow$ AAAATCCCATGGACAGAGGAGCTTG
- CATTLE$\longrightarrow$ AAAATCCCATGGATAGAGGAGCCTG

As a first thing we need to create the frequency matrix, that is basically a matrix in which on the row we have the nucledotides in the sequence (A, T, C, G) and on the columns the number of nucleodites in the sequences. To fill the matrix we simply count the number of apperances of each nucleotide in the sequences for that position and we sum in each cell the value of the pseudocount (in this case B=1)

	1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°	21°	22°	23°	24°	25°
A	5	4	3	5	1	2	2	2	3	1	1	3	3	2	3	2	4	1	2	3	2	1	1	1	1
T	1	2	1	2	4	2	2	1	1	5	1	1	1	3	2	1	1	2	2	1	1	2	3	4	1
C	2	1	4	1	2	4	3	4	3	2	3	2	2	2	1	1	1	1	2	4	2	3	3	1	1
G	1	2	1	1	2	1	2	2	2	1	4	3	3	2	3	5	3	5	3	1	4	3	2	3	6
Total	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9	9

Table 1

In the same way we can evaluate the matrix of the relative frequencies, just dividing the number in each cell for the total of the column:

	1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°	21°	22°	23°	24°	25°
A	0.55	0.44	0.33	0.55	0.11	0.22	0.22	0.22	0.33	0.11	0.11	0.33	0.33	0.22	0.33	0.22	0.44	0.11	0.22	0.33	0.22	0.11	0.11	0.11	0.11
T	0.11	0.22	0.11	0.22	0.44	0.22	0.22	0.11	0.11	0.55	0.11	0.11	0.11	0.33	0.22	0.11	0.11	0.22	0.22	0.11	0.11	0.22	0.33	0.44	0.11
C	0.22	0.11	0.44	0.11	0.22	0.44	0.33	0.44	0.33	0.22	0.33	0.22	0.22	0.22	0.11	0.11	0.11	0.11	0.22	0.44	0.22	0.33	0.33	0.11	0.11
G	0.11	0.22	0.11	0.11	0.22	0.11	0.22	0.22	0.22	0.11	0.44	0.33	0.33	0.22	0.33	0.55	0.33	0.55	0.33	0.11	0.44	0.33	0.22	0.33	0.66
Total	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1	1

Table 2

We can derive the PSSM matrix just by evaluating in each cell the PSSM score, that is:

$$S(pos, nuc) = \log_2 \left[\frac{P(pos, nuc)}{P_r(nuc)} \right]$$

where $P(pos, nuc)$ si basically the value in the cell of the matrix of the relative frequencies and $P_r(nuc)$ is the Random Freq. of Nucleotide, that in this problem is 25% for each nucleotide.

For example in the first cell is:

$$S(pos, nuc) = \log_2[\frac{0.55}{0.25}] = 1.14$$

We can do that for every cell so that we ends up with the following PSSM matrix:

	1°	2°	3°	4°	5°	6°	7°	8°	9°	10°	11°	12°	13°	14°	15°	16°	17°	18°	19°	20°	21°	22°	23°	24°	25°
A	1.14	0.82	0.4	1.14	-1.18	-0.18	-0.18	-0.18	0.4	-1.18	-1.18	0.4	0.4	-0.18	0.4	-0.18	0.82	-1.18	-0.18	0.4	-0.18	-1.18	-1.18	-1.18	-1.18
T	-1.18	-0.18	-1.18	-0.18	0.82	-0.18	-0.18	-1.18	-1.18	1.14	-1.18	-1.18	-1.18	0.4	-0.18	-1.18	-1.18	-0.18	-0.18	-1.18	-1.18	-0.18	0.4	0.82	-1.18
C	-0.18	-1.18	0.82	-1.18	-0.18	0.82	0.4	0.82	0.4	-0.18	0.4	-0.18	-0.18	-0.18	-1.18	-1.18	-1.18	-1.18	-0.18	0.82	-0.18	0.4	0.4	-1.18	-1.18
G	-1.18	-0.18	-1.18	-1.18	-0.18	-1.18	-0.18	-0.18	-0.18	-1.18	0.82	0.4	0.4	-0.18	0.4	1.14	0.4	1.14	0.4	-1.18	0.82	0.4	-0.18	0.4	1.4

Table 3

2.2 Point B

Now given the following two sequences:

- CHIMPANZEE CACGCCTCCCGGTCAGAGACAAGAG
- HOMO SAPIENS CAAGCCTCCCGGTCAGAGACAAGAG

to evaluate for each sequences (chimpanzee and homo sapiens) how likely are they really a version of this gene rather than just a random sequence.
To do that we select in the matrix the value that is associated for each position to the nucleotide present in the sequence and we sum all the values.
For example since the chimpanzee have have in the first position "C" we'll select from the first column -0.18, then in the secon position we'll select (since we have A) 0.82 and so on..
So for the first sequence (the one of the chimpanzee) we have:

$$-0.18+0.82+0.82-1.18-0.18+0.82-0.18+0.82+0.40-1.18+0.82-1.18-0.18-0.18+0.40-0.18+$$

$$+0.40-0.18+0.40+0.82-0.18+0.40-0.18+0.40+1.40=3.56$$

So is $2^{3.56}$, or equally 11.7 times more likely than random probability to be part of the group. The two genes are definitely related.

For the second sequence

$$-0.18+0.82+0.40-1.18-0.18+0.82-0.18+0.82+0.40-1.18+0.82-1.18-0.18-0.18+0.40-0.18+$$

$$+0.40-0.18+0.40+0.82-0.18+0.40-0.18+0.40+1.40=3.14$$

Also for the homo sapiens gene there is definitely some relation, in fact is $2^{3.14}$, or equally 8.8 times more likely than random probability to be part of the group